ARGUS — One-Page AI Deployment & Verification Log

Event: UT Dallas JSOM AI in Finance Case Competition (Fall 2026) | **Proposed Firm:** Charles Schwab Investment Research Team (Westlake, TX — *Independent Prototype*) | **Subject Analyzed:** NVIDIA Corp. (NASDAQ: NVDA, CIK: 0001045810)

1. Core AI Tasks, Tools Used & Architectural Division of Labor

Task Component	AI Tool / Method Used	Why Chosen & Role in ARGUS Architecture
Filing Cross-Check	AI Search & SEC EDGAR Retrieval	Locates verbatim quotes and GAAP/Non-GAAP tables across NVIDIA Q1 FY26–Q2 FY27 releases and the FY2026 Form 10-K (`nvda-20260125.htm`).
Semantic Matching (Core AI Task)	ARGUS Concept-Aware Semantic Engine (`argus_engine.py`)	Links paraphrased claims (e.g., *"compute is revenue"*) to accounting lines (*Accounts Receivable / OCF*) where keyword search scores `0.00`.
Verification & Rules	Deterministic Python & Unit Tests (`test_argus.py`)	Executes 100% of YoY/QoQ %, basis-point margins, DSO math, and 4-tier classification rules so LLMs never hallucinate financial math.

2. Key Prompts, Inputs & Outputs

- **Prompt 1 (Cross-Document Non-GAAP Restatement):** *"Cross-check NVIDIA's Q4 FY26 (Feb 25, 2026) and Q1 FY27 (May 20, 2026) reconciliation tables for methodology changes and historical restatements."* **Output:** Found that starting Q1 FY27, NVIDIA stopped excluding Stock-Based Compensation (SBC) from Non-GAAP measures and restated prior periods—shifting Q4 FY26 Non-GAAP OpEx from **$5,102M** (in Q4 FY26 release) to **$6,666M** (in Q1 FY27 table) and Non-GAAP EPS from **$1.62** to **$1.59**.
- **Prompt 2 (Narrative-vs-Cash-Flow Divergence):** *"Compare CEO Jensen Huang's Q2 FY27 statement ('Its tokens are productive and profitable. Now, compute is revenue') against Q2 FY27 cash flow and receivables trends."* **Output:** Flagged `CMP-NVDA-003`: GAAP Revenue rose **+17.90% QoQ** ($81,615M to $96,221M), but Operating Cash Flow dropped **-52.18% QoQ** ($50,344M to $24,077M) and Free Cash Flow fell **-56.05% QoQ** ($48,554M to $21,341M) as Accounts Receivable surged **+$22,349M (+54.90% QoQ)** to $63,059M (DSO rising from 45.39 to 59.64 days).

3. Verification Checks Performed Against Primary Filings

- **Check 1 — Q4 FY26 SBC Restatement Math (`CMP-NVDA-002`):** Audited Note (B) of Q4 FY26 reconciliation: Total Q4 SBC was $1,633M ($69M Cost of Revenue + $1,217M R&D + $347M SG&A). OpEx-only SBC = `$1,217M + $347M = $1,564M`. Python check: `$5,102M + $1,564M = $6,666M` (exact $0 variance).
- **Check 2 — FY26 OpEx Guidance vs. 10-K Realization (`CMP-NVDA-001`):** Q2 FY26 guided FY26 OpEx growth to the *"high-30% range"* (36.0%–39.9%). Audited FY26 10-K: GAAP OpEx rose from $16,405M to $23,076M (`+40.66%` exact; 41% table) and Non-GAAP OpEx rose from $11,716M to $16,694M (`+42.49%` exact; 42% table), confirming a *Verified numerical mismatch*.
- **Check 3 — Unit-Test Edge-Case Rounding Audit (`tests/test_argus.py`):** For Q2 FY27 OCF QoQ change (`(24077 - 50344) / 50344 = -52.175035%`), unit tests caught a `0.01%` truncation difference between `-52.17%` and standard 2-decimal rounding (`-52.18%`); standardized all assertions to `-52.18%`.

4. What Was Accepted, Modified, or Rejected (12-Pair Labeled Benchmark)

Matching Engine Evaluated (`evaluation_dataset.json`)	Precision	Recall	F1 Score	False Positives
ARGUS Concept-Aware Semantic Engine (Accepted)	**100.0%** (8/8)	**100.0%** (8/8)	**100.0%**	**0 / 4** Negatives
Exact Keyword Overlap Baseline (Rejected as Primary)	33.3% (1/3)	12.5% (1/8)	18.2%	2 / 4 Negatives

Why Keyword Baseline Was Rejected & Guardrails Applied: Exact keyword matching missed 7 of 8 paraphrased positive links while triggering 2 False Positives (`0.925` and `0.969` scores) on unrelated segments (*Automotive* vs. *ProViz* and *Gaming*) sharing boilerplate sentence templates. We replaced naive keyword matching with concept-aware semantic matching + entity conflict penalties, and paired every divergence with mandatory *Contrary Evidence* and *Alternative Explanations*.